

Is This Relation a Function?

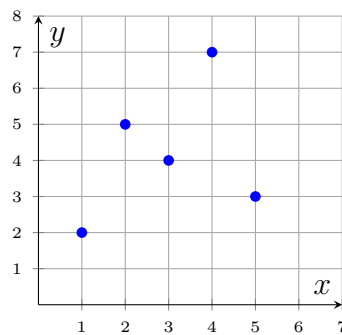
One input, one output: tables, lists of pairs, graphs, and rules

Directions

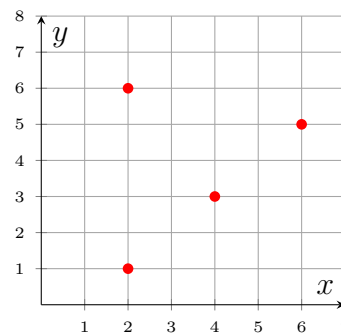
- A relation is a *function* when every input is paired with exactly one output. The rule is about inputs only — outputs are allowed to repeat.
- In a table or a list of pairs, hunt for an input that appears twice with different outputs. On a graph, look for two points sitting directly above each other.
- When a problem asks you to decide, give the reason as well as the decision.

Graph Bank

Graph P



Graph Q



Two relations plotted as sets of points. Only the problems that name graph P or graph Q use this figure.

1. A relation is shown in the table.

Input	1	2	3	4
Output	5	7	9	11

- (a) What output is paired with the input 3?

Answer: _____

- (b) Is this relation a function? Explain how the table shows it.

Answer: _____

2. A relation is the list of pairs $(5, 5)$, $(4, 9)$, $(5, 8)$, $(6, 11)$.

(a) What output is paired with the input 6?

Answer: _____

(b) One input appears in two pairs. Compare its two outputs, in the order they are listed, using $>$, $<$, or $=$.

Answer: _____

(c) Is this relation a function? Explain your decision.

Answer: _____

3. Look at graph P in the Graph Bank.

(a) Write the y -value of the plotted point whose x -value is 4.

Answer: _____

(b) Is the relation shown by graph P a function? Explain how the picture shows it.

Answer: _____

4. A relation is given by the rule $y = 4x - 3$, where x is the input and y the output. Find the output when the input is 5.

Answer: _____

5. A relation is shown in the table. Notice that one input appears twice.

Input	8	6	8	9
Output	12	5	8	5

(a) What output is paired with the input 9?

Answer: _____

(b) Compare the two outputs of the repeated input, in the order the table lists them, using $>$, $<$, or $=$.

Answer: _____

(c) Is this relation a function? Explain your decision.

Answer: _____

6. A relation is given by the rule $y = 2x + 1$.

(a) Find the output when the input is 7.

Answer: _____

(b) Find the output when the input is 12.

Answer: _____

7. Look at graph Q in the Graph Bank.

(a) Two of its plotted points sit directly above each other. Compare their y -values, the lower point first, using $>$, $<$, or $=$.

Answer: _____

(b) Is the relation shown by graph Q a function? Explain how the picture shows it.

Answer: _____

8. A relation is the list of pairs $(1, 4)$, $(2, 4)$, $(3, 4)$, $(4, 4)$.

(a) What output is paired with the input 3?

Answer: _____

(b) The output 4 is used every time. Is this relation still a function? Explain your decision.

Answer: _____

9. Relation R is the table below. Relation S is the list of pairs $(4, 3)$, $(6, 12)$, $(4, 11)$, $(8, 12)$.

R input	1	2	3	4
R output	6	8	10	12

(a) What output does R pair with the input 4?

Answer: _____

(b) In S , one input appears twice. Compare its two outputs, in the order they are listed, using $>$, $<$, or $=$.

Answer: _____

(c) Which of R and S is a function? Explain your decision for each one.

Answer: _____

10. A relation is given by the rule $y = 3x + 5$.

(a) Find the output when the input is 6.

Answer: _____

(b) Explain why a relation given by a rule is always a function, while a relation given as a list of pairs might not be.

Answer: _____